

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY – PHASE I (Analysis)

Problem Identification & Class Mapping (CO1)

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1. Problem Title

Contact Management System

2. Problem Description

A Contact Management System is used to store and manage personal and business contacts digitally. The system allows users to add, update, delete, and search contacts easily. Each contact contains information such as name, phone number, email, address, birthday, and company. The system helps organize contacts efficiently and allows quick retrieval of information.

The implemented system stores all contact data persistently in a flat file (contacts.txt) using Java File Handling. Login and logout events are automatically recorded with timestamps in log.txt. Category assignments are saved to categories.txt, enabling grouped contact management.

3. Identified Entities (Classes)

Class Name	Description
User	Person who uses the system to manage contacts. Handles login/logout with timestamped activity logging.
Contact	Stores details of a person such as name, phone number, email, address, birthday, and company. Serializes itself to a comma-separated string for file storage.
ContactManager	Handles all CRUD operations. Reads from and writes to contacts.txt using FileReader and FileWriter.
Category	Used to group contacts into types like personal or business. Saves category data to categories.txt.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods (Behaviours)
User	userId, username, password, email	login(), logout(), validateCredentials()
Contact	contactId, name, phoneNumber, email, address, birthday, company	toFileString(), display(), getters, setters
ContactManager	FILE_NAME (contacts.txt)	addContact(), searchContact(), deleteContact(), displayContacts(), viewContact(), updateContact(), loadFromFile(), saveToFile(), getNextId()
Category	categoryId, categoryName	createCategory(), assignCategory()

5. Relationships Between Classes

From	Relationship	To / Description
User	Uses	ContactManager — User calls CRUD operations via menu
ContactManager	Manages	Contact — loads, stores, updates, and deletes Contact objects
Contact	Serialized to	contacts.txt — one contact per line in CSV format
Category	Groups	Contact — assigns contacts to named categories, saved to categories.txt
User	Logs to	log.txt — every login/logout recorded with timestamp

6. File Handling Design

The system uses three persistent files:

File	Managed By	Purpose
contacts.txt	ContactManager	Stores all contact records in CSV format. Append on add; overwrite on update/delete.
log.txt	User	Records every login and logout with date and time using LocalDateTime.

categories.txt	Category	Stores category assignments. Appended whenever a contact is assigned to a category.
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